

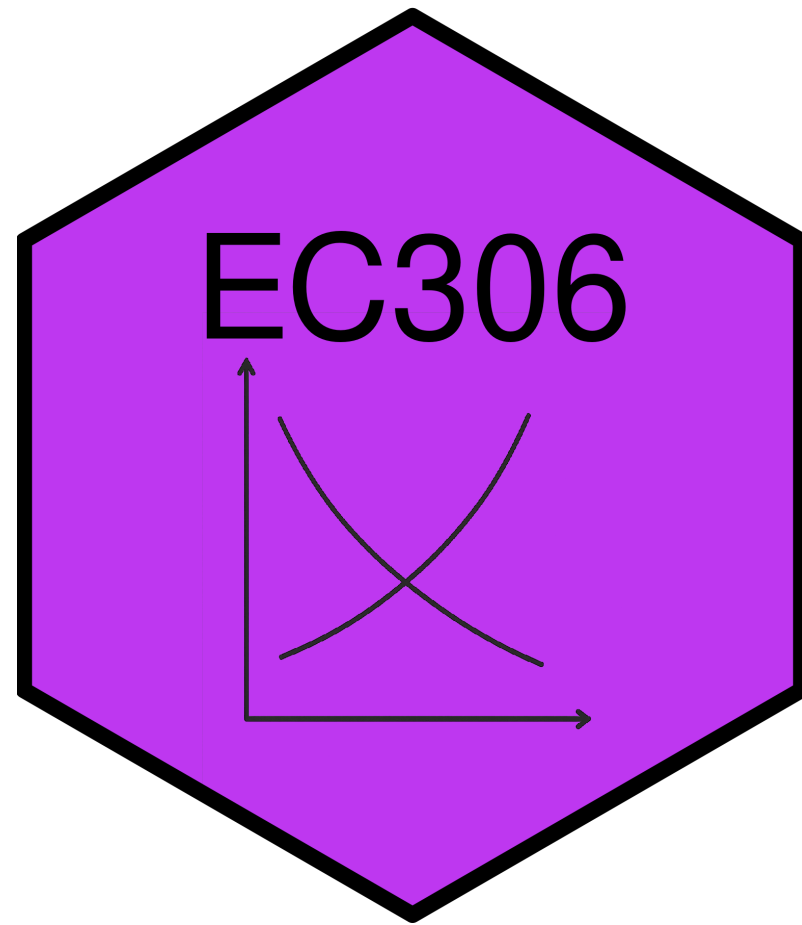
Wages and Employment in a Single Labour Market

EC306- Wages and Employment

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Goals of This Section

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- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

Introduction

Introduction

- We have analyzed labour demand and supply separately in detail
- The equilibrium wage and employment is determined by the interaction of the two
- Market power plays an important role
 - In both goods markets and labour markets
- With a full understanding on how wages and employment are determined, we can analyze policies
- The minimum wage is one of the most prominent labour market policies
 - In theory, it leads to unemployment
 - The evidence is more mixed

Competitive Firms

Introduction

- We start with firms that are competitive in the goods and labour markets
- Perfect competition in both implies that
 - Firms take the market wage as given
 - They take the market price as given
 - Competition means they are too small to influence either
- We analyze the equilibrium wage and employment for a homogeneous type of labour
 - All workers are the same

Market Demand and Supply

- The market demand and supply are the sum of the individual firms' demand and supply at each wage
- Here we determine the demand and supply curve for a single labour market
 - Could be an specific occupation in a specific region
 - Example: electricians in Ontario, professors in Canada, etc.
 - How the market is defined depends on context
 - But it is for a single type of labour

Market Demand and Supply

- Example is two firms with different elasticities of demand
 - Each firm demand curve is their value of marginal product
 - Firms may face different product prices if they sell different things (but hire the same labour)
- Each firm faces the same wage in the labour market
 - They perceive the supply curve as horizontal in the long run
 - They take the wage as given and can get as much labour as they want at that wage
 - Market supply is upward sloping
- Market demand is horizontal sum of individual demands
 - Add up the quantities demanded at each wage for each firm
 - Repeat for all wages

Market Demand and Supply

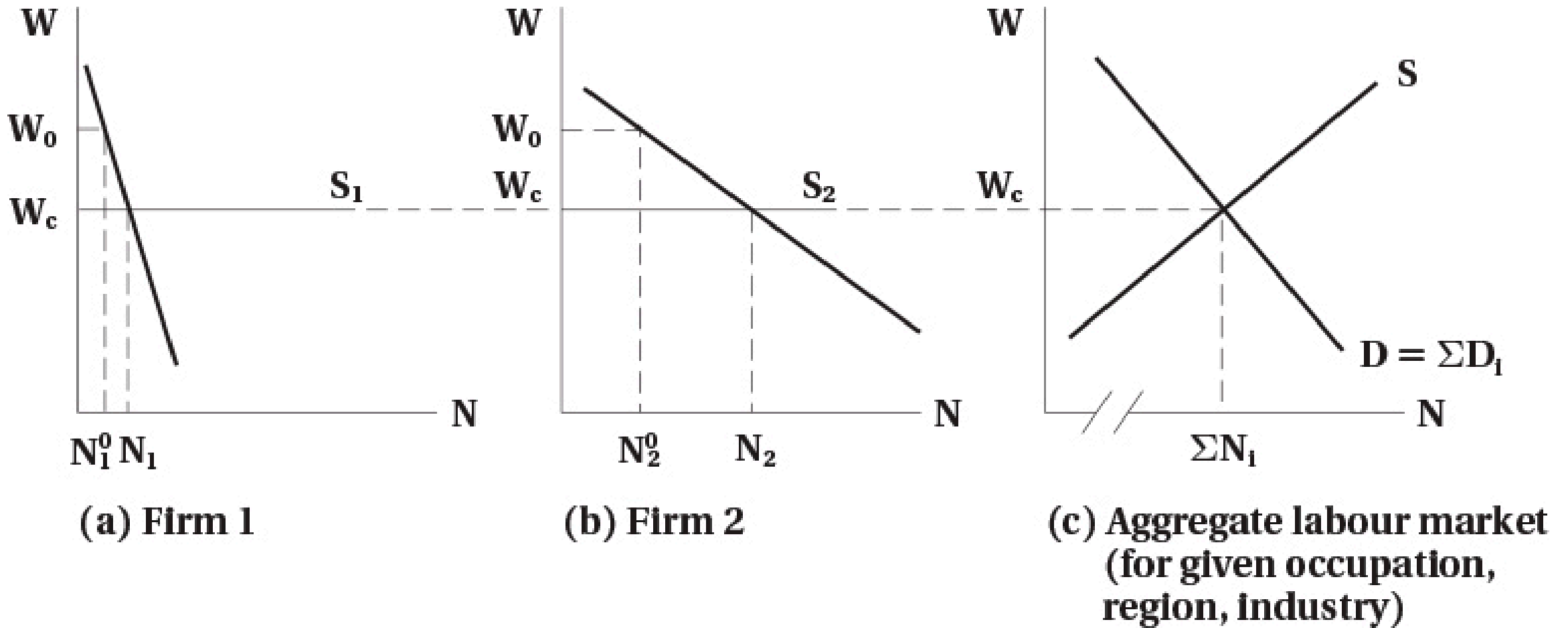


FIGURE 7.1 Competitive Product and Labour Markets

Firm Demand and Supply in the Short Run

- Above we analyzed the longer run
 - Firms can get as much labour as they want at the going wage
- In the short run, firms may face an upward sloping supply curve
 - They have to pay higher wages to attract more workers
 - This could be because of contracts, or other frictions
- If demand rises, they would pay higher wages to attract workers
- But because wages are high, over the long term more workers enter the market
 - Short run supply shifts out
 - Wage falls back down to long run level

Firm Demand and Supply in the Short Run

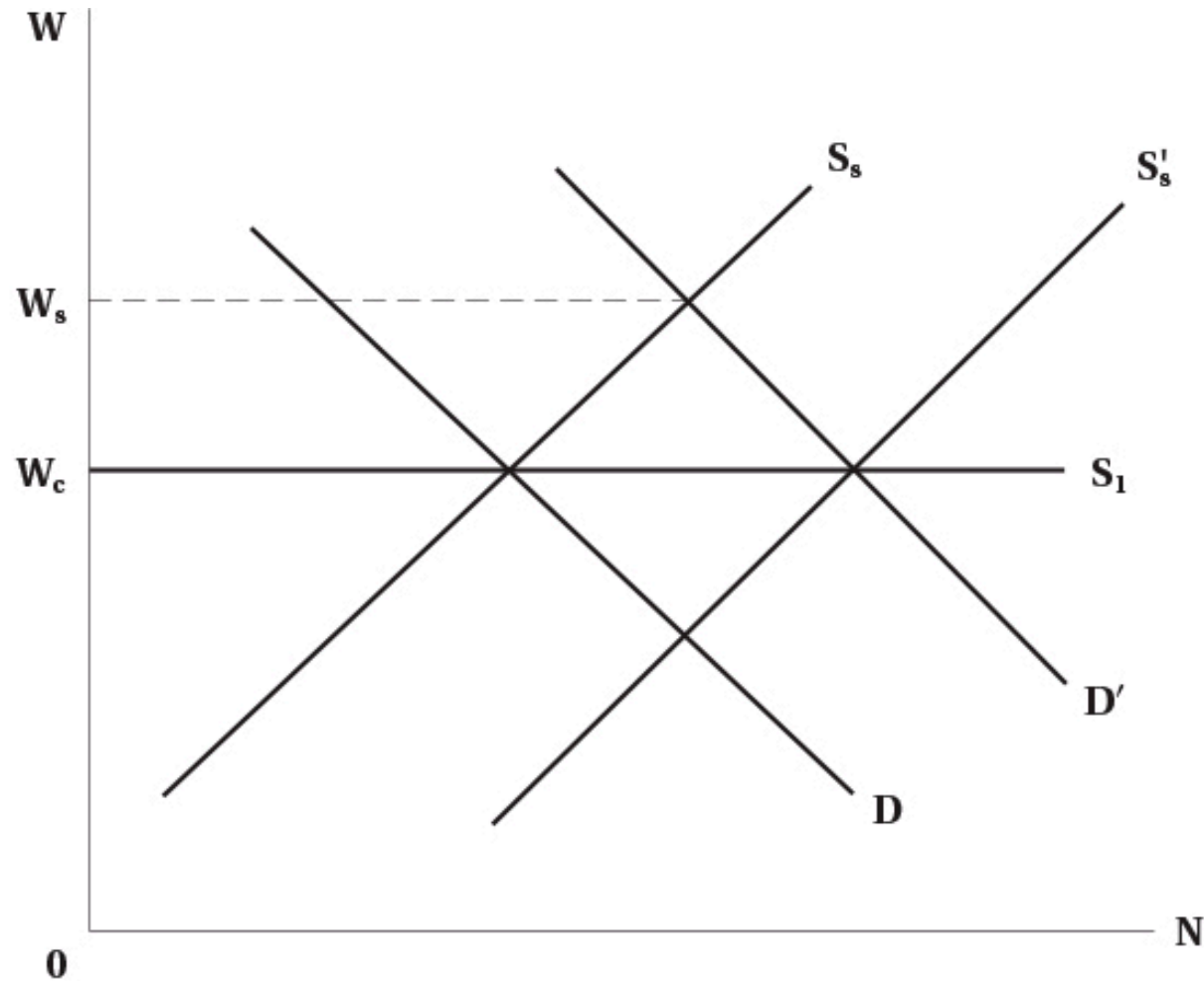


FIGURE 7.2 Short-Run and Long-Run Labour Supply Schedules to the Firm

Imperfect Competition in the Product Market

Monopoly

- In competitive markets there are many sellers of a product
- In a monopoly, there is only one
- This gives the firm market power
 - The ability to set prices above marginal cost
- For a monopolist, the demand curve is where wage equals marginal revenue product

$$w = MR \times MP P_N$$

- For a monopolist, marginal revenue is less than price
 - Because to sell more, they have to lower price on all units sold
 - So marginal revenue product is less than value of marginal product (VMP)

Monopoly

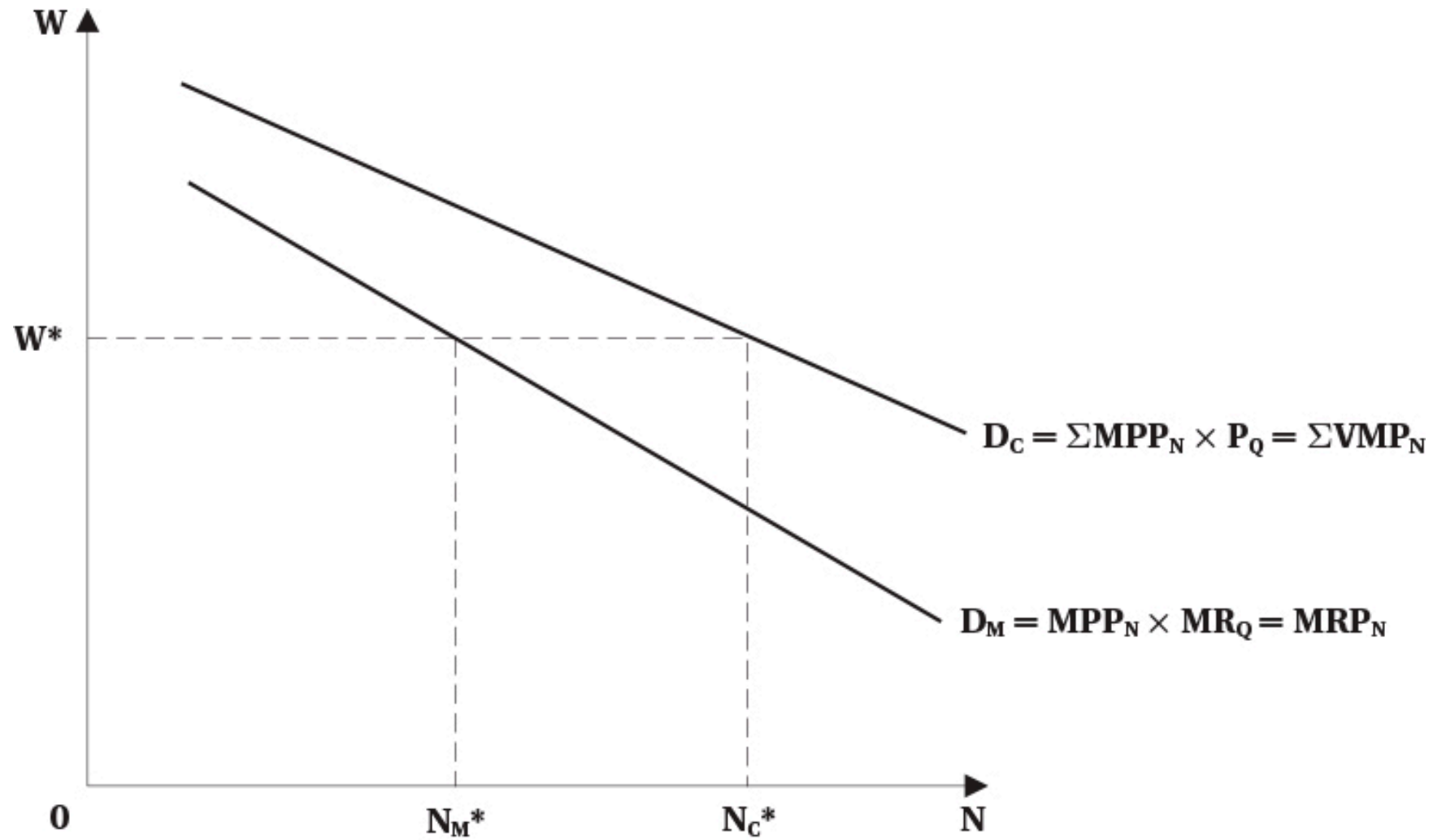


FIGURE 7.3 Monopolist versus Competitive Demand for Labour

Monopoly

- Because monopolists have market power
 - They can raise the price of their product above competitive level
 - This will decrease output
 - They demand less labour
- Monopolists therefore will hire fewer workers
 - Compared to a competitive firm in the same market
- Note that power in the product market does not mean power in the labour market
 - The firm is still one of many hiring workers in a particular labour market

Monopolistic Competition and Oligopoly

- Many markets are neither perfectly competitive nor pure monopoly
- Monopolistic competition
 - Many firms selling differentiated products
 - Each firm has some market power because of product differentiation
 - Can set prices above marginal cost
- Oligopoly
 - A few firms dominate the market
 - Each firm has significant market power
 - But must consider reactions of competitors when setting prices
- Demand for labour is still based on marginal revenue product
 - Which is less than value of marginal product
 - So firms hire fewer workers than in competitive markets



Additional Details

- All of the previous analysis assumes that labour markets are perfectly competitive
 - Firms can hire all labour they want at the given wage
- That is not always the case in reality when firms have power in the product market
 - Employees may unionize
 - Firms may hire a large number of the market's workers
- In these cases, firms may not face a perfectly elastic supply curve for labour
- Later we will analyze imperfect competition in the labour market

Math of Labour Market Equilibrium

Introduction

- We know that market equilibrium is where demand and supply are equal
- You can do that graphically or mathematically
- In this section we cover both
- To make the math easier, we use linear demand and supply curves
 - This is just for simplicity
 - The same principles apply with non-linear curves

Demand and Supply Functions

- The demand and supply for labour are functions of wages and other factors
 - Demand: $N^D = f(w, Z)$
 - Z is a list of other factors affecting demand
 - Supply: $N^S = g(w, X)$
 - X is a list of other factors affecting supply
- Z and X are things that shift the curves
 - For example, technology, prices of other inputs, preferences, demographics, etc.
 - We call these **exogenous** variables
 - Determined outside the model
- N^D , N^S , and w are **endogenous**
 - Determined within the model

Demand and Supply Functions

- Assume the demand and supply functions are linear
 - Demand: $N^D = a + bw$
 - a, b are constants
 - Supply: $N^S = c + fw$
 - c, f are constants
- To find equilibrium, set demand equal to supply

$$a + bw^* = c + fw^*$$

Demand and Supply Functions

- Rearranging to solve for equilibrium wage:

$$w^* = \frac{c - a}{b - f}$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af}{b - f}$$

- Use these equations to analyze how changes in exogenous variables

- Changing a would shift the demand curve

- Example: increase in product price

- Changing c would shift the supply curve

- Example: increase in population



Demand and Supply Functions

- To draw the demand and supply curves, we need w on the left side
- Rearranging the demand function:

$$w = \frac{N^D - a}{b}$$

- Rearranging the supply function:

$$w = \frac{N^S - c}{f}$$

- Next slide plots these functions

Demand and Supply Functions

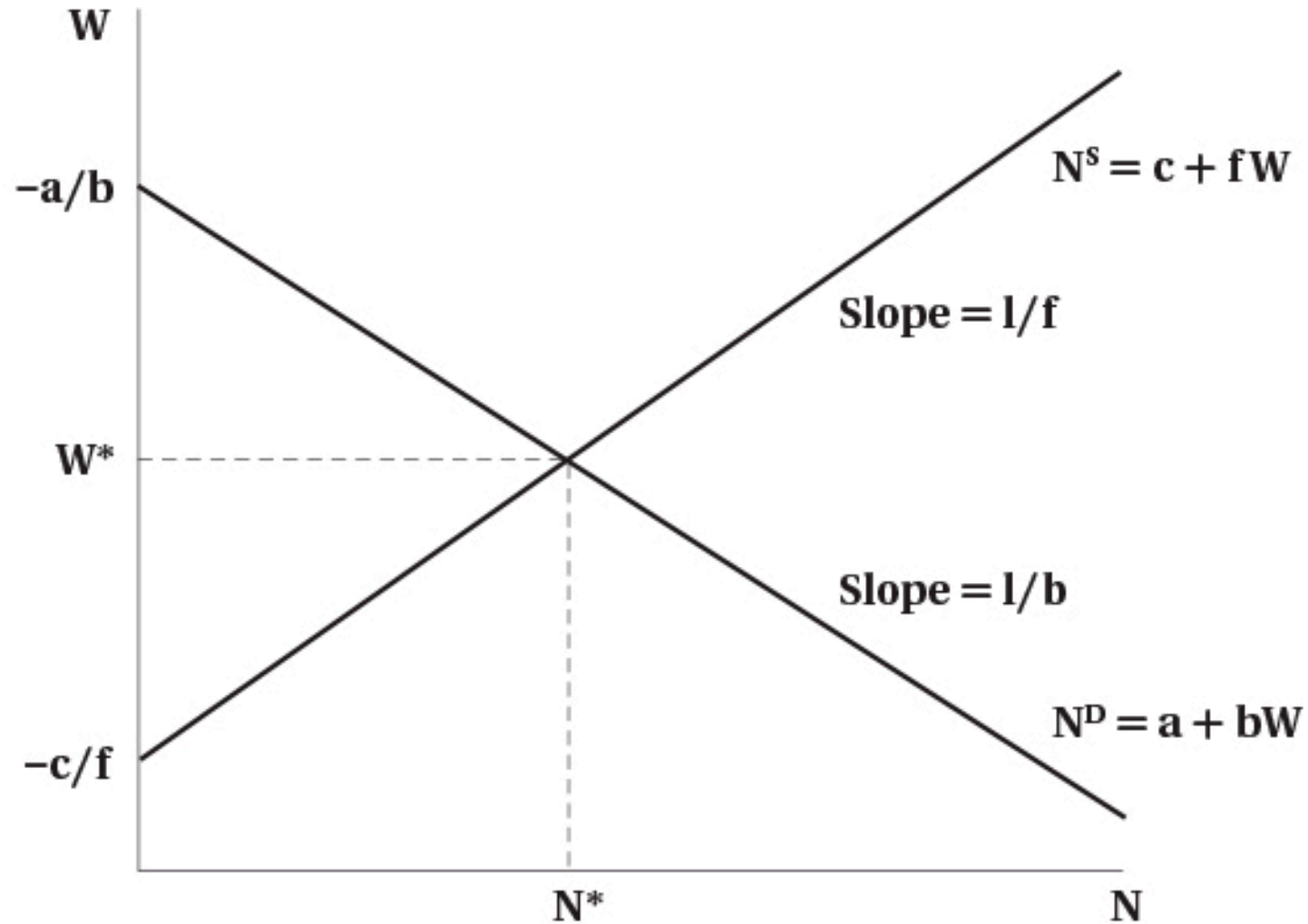


FIGURE 7.4 Linear Supply and Demand Functions

Payroll Tax

- We can use this model to analyze various policies
- One common policy is a payroll tax on employers
 - Examples: Canada Pension Plan (CPP), Employment Insurance (EI)
- These are typically a percentage of wages paid by employers
- But we will analyze a fixed amount per hour for simplicity
 - Suppose the tax is T per hour
- This drives a “wedge” between the wage paid by employers and the wage received by workers in equilibrium

$$w_r = w_e - T$$

Payroll Tax

- The demand function is:

$$w_e = \frac{N^D - a}{b}$$

- The supply function is:

$$w_r = \frac{N^S - c}{f}$$

Payroll Tax

- Substitute into the equilibrium price equation and set quantity demanded equal to quantity supplied:

$$\frac{N^* - c}{f} = \frac{N^* - a}{b} - T$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af - bfT}{b - f}$$

Payroll Tax

- The equilibrium wage received by workers is

$$w_r^* = \frac{c - a}{b - f} - \frac{b}{b - f} T$$

- The equilibrium wage paid by employers is

$$w_e^* = \frac{c - a}{b - f} - \frac{f}{b - f} T$$

Payroll Tax

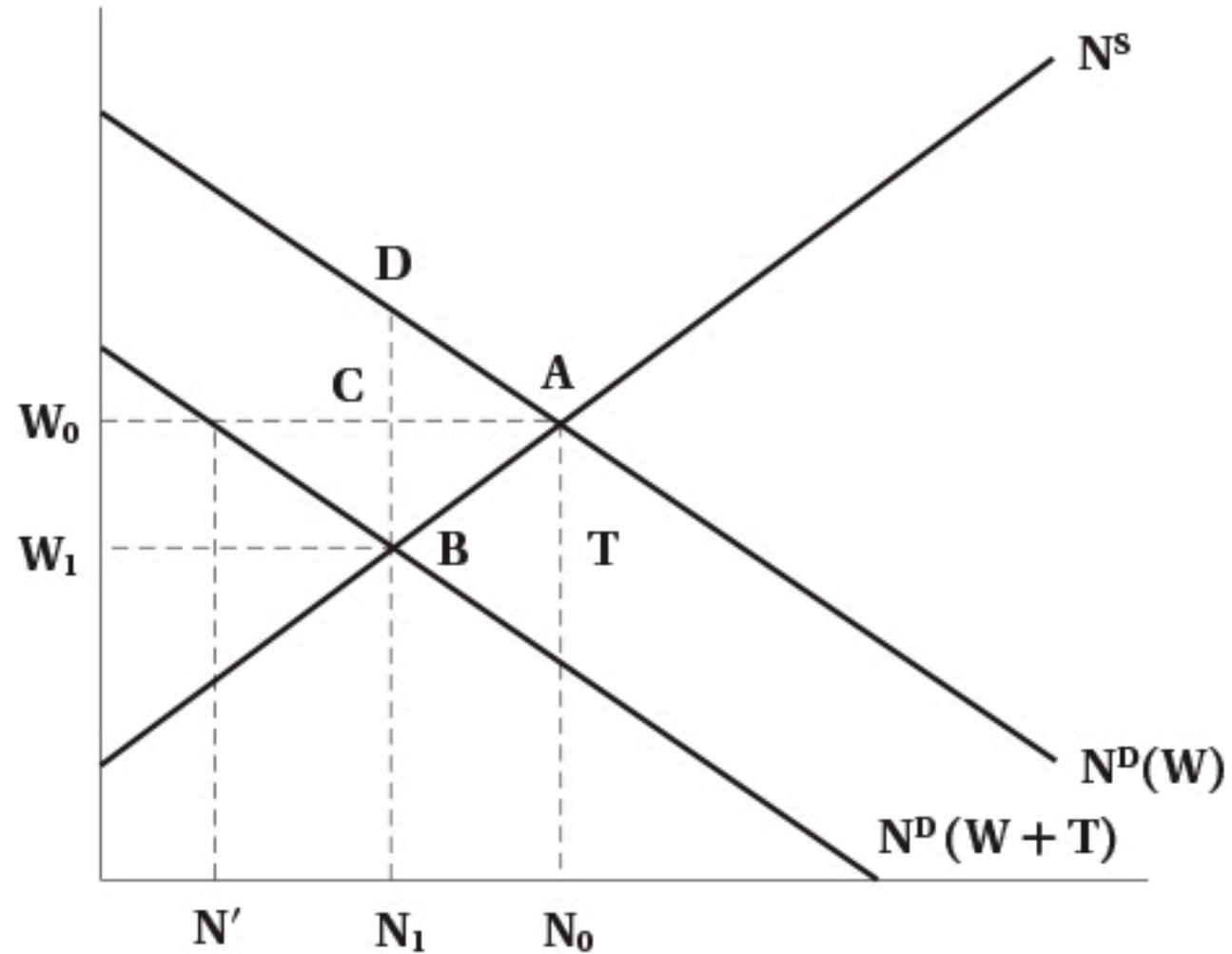


FIGURE 7.5 The Effect of a Payroll Tax on Employment and Wages

Payroll Tax

- The previous slide highlights the difference between the statutory and economic incidence of a tax
 - Statutory incidence: who is legally responsible for paying the tax
 - In this case, firms
 - Economic incidence: who actually bears the burden of the tax
 - Shared between firms and workers
 - Workers receive a lower wage than before
 - Firms pay a higher one
- Economic incidence depends on elasticities of supply and demand
 - More elastic supply means workers bear less of the tax burden
 - More elastic demand means firms bear less of the tax burden

Evidence

- Theory predicts lower employment from a payroll tax
- Also predicts economic incidence may be shared between firms and workers
- Evidence based on empirical studies is
 - In the long run, workers bear most of the burden of the tax through lower wages
 - Employment effects are small